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The Great Transition of Work: Redefining Vocational Education and Training

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General Purpose Technologies through History



- **Electricity (1880s)**
  - Powered factories, lighting and appliances
  - Reorganised work around the assembly line



- **The Computer (1950s-80s)**
  - Automated calculation, records and control
  - Created entirely new categories of work



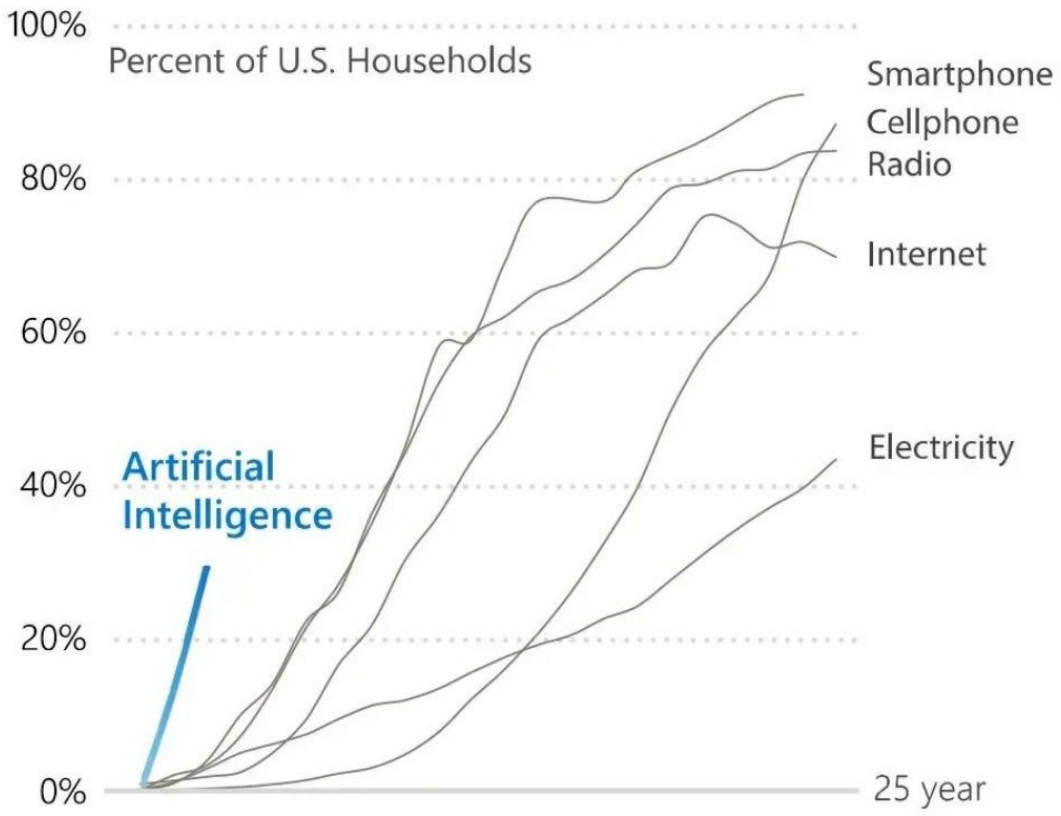
- **The Internet (1990s)**
  - Collapsed the cost of communication and coordination
  - Enabled global, distributed and remote work



- **Artificial Intelligence (2020s)**
  - Automates cognitive and creative tasks
  - Reshapes skills, jobs and how we learn

What's different this time?

Speed of Diffusion



Source: Microsoft AI Economy Institute (2025)

Time taken to reach 100 million users

Technology / Platform	Time to 100M users
Telephone	75 years
Mobile phone	16 years
World Wide Web	7 years
Facebook	4.5 years
Instagram	2.5 years
TikTok	9 months
ChatGPT	**2 months**

Sources: European Commission (2020); UBS analysis, reported by Reuters (2023)

An Existential Question

> *If AI can increasingly perform the tasks for which we train people, what is vocational education for?*

AI affects

What to learn	How to learn

Not less VET, but different VET

Three Shifts

FROM

Jobs

⬇

Tasks & redesigned roles

FROM

Task mastery

⬇

Adaptive expertise

FROM

Front-loaded training

⬇

Lifelong learning

What to learn

How & when to learn

Impact of AI on Jobs

New jobs will be created or enabled even as some jobs disappear

~ 60% of US workers in 2018 employed in jobs that didn't exist in 1940 (Autor et al. 2024)

- **World Economic Forum Future of Jobs Report** 2025 projections:
  - 170 million new roles and 92 million displaced by 2030
  - **39% of workers' core skills expected to change**
  - 63% of employers naming skills gaps as the leading barrier to transformation

Impact on tasks and jobs will be uneven, depending on speed of AI adoption / technology diffusion

The Risks

Certain jobs or tasks could be displaced

- **Uneven impact** across jobs/sectors – substitution vs output
- Job displacement also depends on speed of AI adoption or technology diffusion
- Some evidence of dip in **entry-level jobs**
    - E.g. Brynjolfsson et al. 2025, IAL-Burning Glass Institute study

If AI takes over entry-level tasks, how will new workers acquire the experience needed to become experts?

Commoditisation of skills and potential loss of wage premium

- Historical example: GPS and London Cab drivers

The Opportunities

Increasing productivity and augmenting human capability

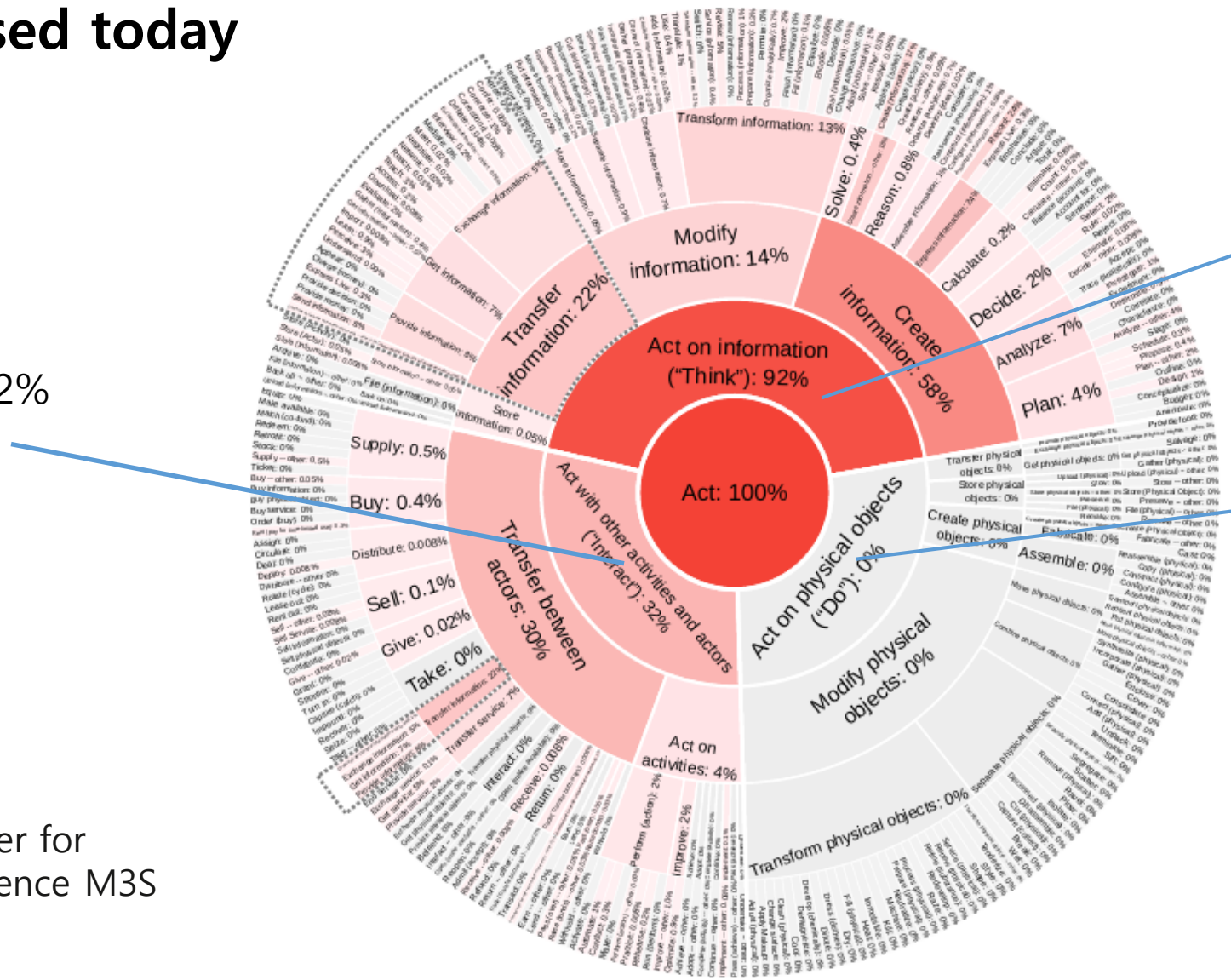
- Increase efficiency and productivity by taking over routine work
- Enable “one person companies”

Enable workers to take on higher professional responsibilities, blurring lines between traditional occupational boundaries

- e.g. technicians, nurses

From Jobs to Tasks

Where AI is used today



Interact: 32%

Act on Information (Think): 92%

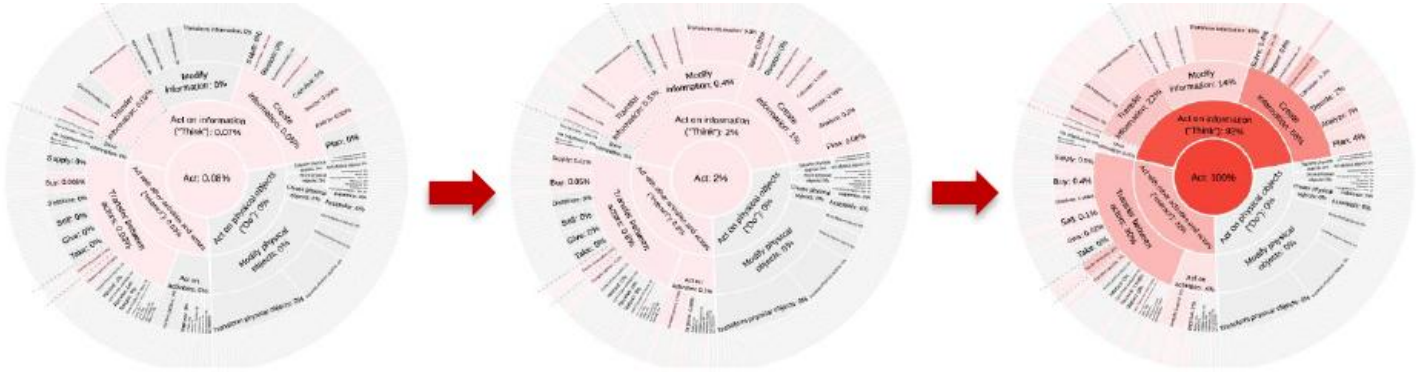
Act on Physical Objects (Do): 0%

Source: MIT Center for Collective Intelligence M3S (2026)

From Jobs to Tasks

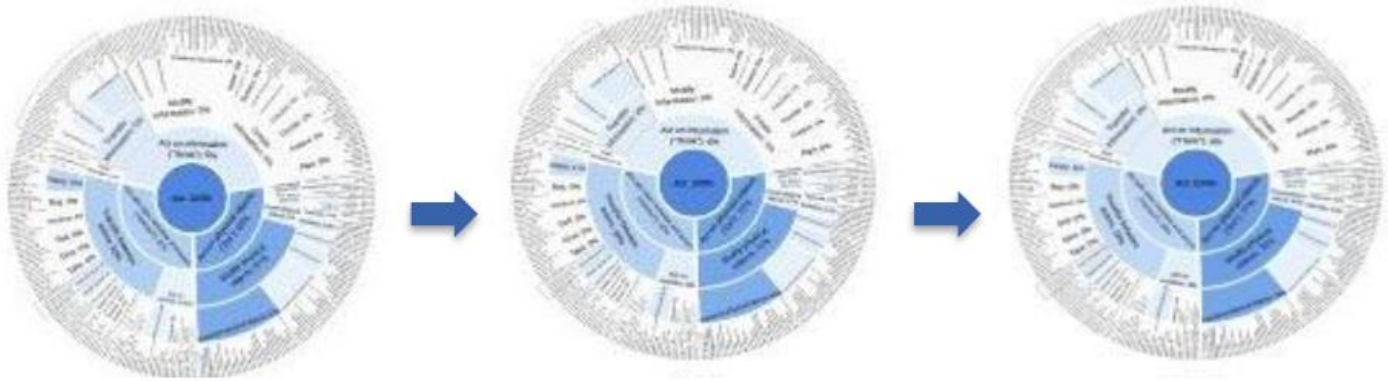
AI vs Robotics - While AI applications have rapidly expanded across work activities, robotic systems have been relatively stable and concentrated in a few activities

AI software applications



2016 2020 2025

Robotic systems



2015 2020 2024

Source: MIT Center for Collective Intelligence M3S, via Kathleen Kennedy (2026)

Redeploying Tasks

Every task in a role is routed to one of four lanes — deciding what AI does, what humans do, and what stops being done.

	Route	Criteria	Impact
Lane 1	Full Automation	Routine, rule-based, data available and compliant.	Shifts execution fully to AI; stil requires human monitoring & accountability
Lane 2	Human–AI Collaboration	High value, cognitive, varied.	AI provides pattern processing; humans provide judgment and edge-case management.
Lane 3	Human-Led	High relational value, complex judgment, deep tacit knowledge.	Protected and strengthened; often the core value of the role.
Lane 4	Eliminated	No longer serves organizational goals.	Clears space for new value creation.

From Jobs to Tasks

Jobs will need to be deconstructed into tasks and reconstructed

Deconstruct the work

1. Task classification

Classify tasks by repetition, interaction, and physical or mental demands.

2. Return on improved performance

Assess each task's value, risk and potential performance gains.

Redeploy the work

3. Automation Type

Choose how to automate or augment tasks using RPA, social robotics or AI.

4. Work alternatives

Consider shared services, internal talent pools, contractors or outsourcing. Assess risks and trade-offs.

Reconstruct the work

5. Reconstruction

Combine remaining and new tasks into existing or new roles.

Match work to talent

Choose the best source of talent, develop the skills needed and address deployment constraints.

From Task Mastery to Adaptive Expertise

Traditional VET question

“Can you perform this task?”

Future VET question

“Can you perform it, understand it, improve it and adapt when the task changes?”

Technical mastery	Occupational skills and domain knowledge
Human judgement	Contextual decision-making, communication, creativity, collaboration
Adaptive capacity	Learning, problem-solving, technological literacy, ability to move into adjacent roles.

AI may in fact increase the value of vocational learning

AI can increasingly provide knowledge, instructions, diagnosis and procedural guidance. But **real-world work requires more than following instructions**.

Real workplaces contain ambiguity, exceptions, incomplete information and consequences.

Workers must exercise contextual judgement, troubleshoot problems and make decisions with real-world consequences. Hence the enduring value of:

- **tacit knowledge**
- **contextual judgement**
- **troubleshooting**
- **physical and interpersonal competence**
- **responsibility/accountability**

This is especially relevant in healthcare, maintenance, construction, hospitality, advanced manufacturing, logistics, etc.

Technical mastery and domain expertise still matter

AI can augment human performance but does not eliminate the need to develop foundational knowledge and skills.

Domain knowledge is essential to critically evaluate, validate and effectively use AI output.

Just as calculators have not made learning mathematics obsolete, GenAI does not eliminate the need to learn writing, coding or other foundational skills.

Learning ≠ performance - Completing tasks successfully with AI does not necessarily mean developing genuine understanding or competence.

Learners need opportunities for independent thinking, practice and productive struggle to develop expertise.

Critical Core Skills



From Task Mastery to Adaptive Expertise

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Tech skills

Tech skills

Tech skills

Thinking critically

Staying relevant

Staying relevant

Interacting with others

Interacting with others

Thinking critically

Ethics

Future of Jobs Report 2025

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Top 10 fastest growing skills by 2030

1. AI and big data
2. Networks and cybersecurity
3. Technological literacy
4. Creative thinking
5. Resilience, flexibility and agility
6. Curiosity and lifelong learning
7. Leadership and social influence
8. Talent management
9. Analytical thinking
10. Environmental stewardship

Learning Agility

Willingness and **ability** to

- learn new things to obtain results under difficult or different conditions (Lee and Song 2021)
- learn from experience and apply lessons learnt to perform successfully under new or first-time conditions (De Meuse 2010)

Closely related to mental and psychological factors:

- **Individual disposition**
  - Goal orientation, cognitive / metacognitive ability, openness to experience
- **Cognitive processes – habits of mind**
  - Cognitive simulations, counterfactual thinking, pattern recognition
- **Behavioural Processes**
  - Feedback seeking, experimentation, reflection

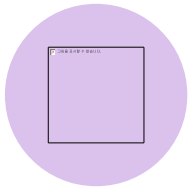
Source: DeRue, Ashford and Myers. Learning Agility: In Search of Conceptual Clarity and Theoretical Grounding. *Industrial and Organizational Psychology*. 2012;5(3):258-279.

How AI can support & accelerate learning

- **Personalised tutoring**
  - 24/7 guidance, adaptive exercises and feedback
  - Supports flipped classrooms and individual learning pathways
- **Research and content development**
  - Helps learners research and synthesise information
  - Assists educators in developing materials and assessments
- **Simulations and experiential learning**
  - Practise realistic workplace scenarios safely
  - Repeated practice with immediate feedback

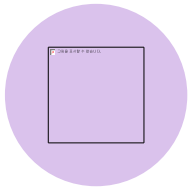
Holistic VET for the AI Age

Institute of Technical Education



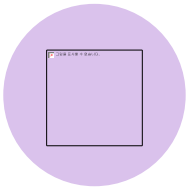
HANDS-ON

Technical competence, experiential learning



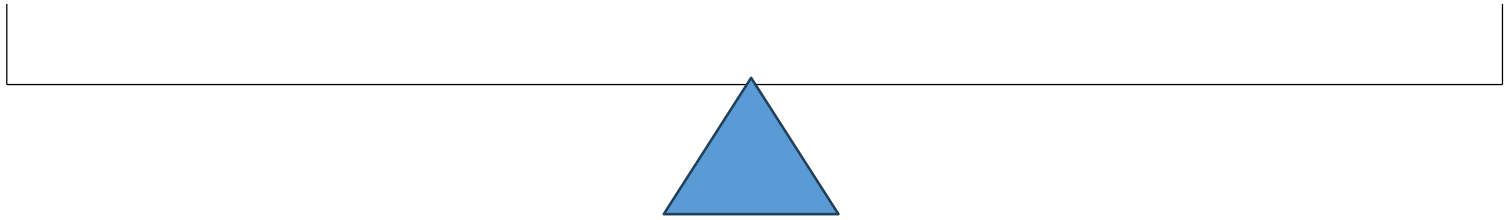
MINDS-ON

Critical thinking, judgement, creativity & innovation



HEARTS-ON

Values, empathy, interpersonal skills



LEARNING AGILITY

How Singapore's Institute of Technical Education is adapting to the AI Age

AI-focused qualifications:

- Introduced nrew course in AI Applications to prepare students for emerging AI-related roles.

AI-enhanced teaching and learning

- Using AI tools and immersive simulations to support personalised and experiential learning.

Industry partnerships:

- SME@AiTE initiative enables students to develop AI solutions for real-world business challenges.

Stronger workplace learning:

- Expanded internships and industry exposure to develop practical skills and adaptability.

Lifelong upskilling:

- Offering short courses in generative AI and its workplace applications for adult learners.

Barriers to Training Take-Up by Individuals

Situational (cost-benefit and constraints)	Cost, work/life constraints & employer limitations Lack of skills recognition by employers
Informational	Lack of clear labour market information Lack confidence in quality of CET
Psychological	Internal attitudes such as fear of failure or low self-concept
Societal	Gender stereotypes, age-related disadvantages, and undervaluation of trades & craft

Source: “Skills-First: What Does It Mean for Me?” Centre for Skills-First Practice Working Paper No. 2 (July 2025)

From Front-loaded Training to Lifelong Learning

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Singapore's SkillsFuture Movement (since 2015)

Challenge 1: Global Trends



Technology and globalisation changing the nature of jobs and fueling rapid pace of skills obsolescence

SKILLSfuture

A national movement to provide Singaporeans with the opportunities to develop to their fullest potential throughout life through skills mastery and lifelong learning.

Challenge 2: Local Labour Market Conditions



Ageing workforce and slowing workforce growth



Aspiration-fuelled "paper chase" could lead to overeducation and underemployment



Barriers to training

- Employers' hesitation
- Individuals do not see the need to upgrade

Singapore's SkillsFuture Movement

"We will build on these foundations to create a new environment for **lifelong learning**. It is critical to our future. It will develop the **skills and mastery** needed to take our economy to the next level. More fundamentally, it aims to **empower each Singaporean to chart their own journey in life,** and gain fulfilment at work and even in their senior years.

We have called this development effort 'SkillsFuture'. It marks a **major new phase of investment in our people**, throughout life."

*Then-DPM **Tharman Shanmugaratnam**, Budget Statement 2015*



SkillsFuture Work-Study Programmes

Blend classroom-based learning with structured on-the-job training, enabling individuals to gain industry-relevant experience while working towards nationally recognised qualifications

Certificate (6 weeks to 18 months), **Diploma** (12 – 36 months) and **Post-Diploma** (9 – 18 months) programmes

Programme expansion: From 15 Work-Study Programmes in 2015 to about 150 programmes in 2025, spanning a wide range of industry sectors.

Participation: Since 2015, about 16,000 individuals have participated, with some 4,000 companies benefiting from the programmes.

Outcomes of Work-Study Programmes

Employment within six months after completing the programme:

- More than 9 in 10 graduates were employed
- Around 7 in 10 stayed with the company that sponsored them for the programme.
- Graduates earned on average 33% more than at the start of the programme.

Study of wage outcomes of Work-study Post-Diploma programme (by Ministry of Trade and Industry) :

- Trainees earned more during programme than peers who went straight to work
- Wage premium sustained, averaging about 9% 1-2 years after programme completion and 6% in the subsequent 2 years.

Redefining Vocational Education

What must change in the VET ecosystem?

Training institutions	Employers	Governments	Individuals
• Agile curricula, modular programmes and AI-enabled learning	• Job redesign, workplace learning and opportunities for career mobility	• Skills intelligence, training incentives and support for career transitions	• Learning agility, ownership of skills development and openness to change

The Future of VET: Not less, but different

Technical	Strong foundations in occupational and domain knowledge
Human	Judgement, creativity, problem-solving and collaboration
Adaptive	Learning agility and the ability to take on new roles

From preparing people for a job to preparing them for a lifetime of changing work.